

End-to-End Processing-on-Chip Wearable Ear EEG Device with Tiny Neural Network for Multilevel Stress Detection

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Abstract—The study proposes an innovative method for predicting stress levels that employs electroencephalography (EEG) data collected from behind the ear (BTE) and incorporates tiny neural networks (TNN). The system, which consists of an EEG processing device and a companion smartphone application for monitoring, is designed to: (1) collect EEG signals from behind the ear rather than the scalp, prioritizing comfort and reducing hair interference; (2) filter out irrelevant signals from the desired EEG; (3) extract relative power spectral density (PSD) features from EEG data using Fast Fourier Transform (FFT); and (4) classify stress levels using a proposed TNN. Importantly, all processing stages occur on the chip itself, rather than on external devices such as PCs or servers, providing fast and synchronized processing and preventing data loss during transmission. In addition, (5) a user-friendly smartphone app is being created for rapid and straightforward stress level monitoring. Experimental results show that the on-chip TNN is successful, with accuracy of 90.83%, precision of 90.87%, recall of 90.83%, and an F1-score of 0.9081, all verified using leave-one-subject-out cross-validation (LOOCV). This merging of BTE EEG analysis with on-chip TNNs offers the potential for enhancing mental stress detection and other health-related uses.

Keywords—Multilevel Stress Detection; Behind-the-ear EEG; Tiny Neural Networks; On-chip Processing.

I. INTRODUCTION

Mental stress encompasses emotional or psychological tension experienced when facing challenging situations, adversely affecting both mental and physical health [1]. Common manifestations include depression, anxiety, and sleep disturbances, all of which can impact one's performance, motivation, and focus [2].

Previous researches demonstrated that both positive and negative stress trigger bioelectrical changes in the brain, akin to other neural activities. EEG offers a non-invasive means to record brain activity, allowing for the assessment of mental stress levels by analyzing brain wave patterns [3]. These patterns are reflected in the electrical signals detected by electrodes placed on the scalp, representing the collective neural activity.

Previous studies have employed various machine learning algorithms to analyze EEG signals for classifying mental stress versus relaxation states [4], [5], [6]. One approach utilized a support vector machine (SVM)

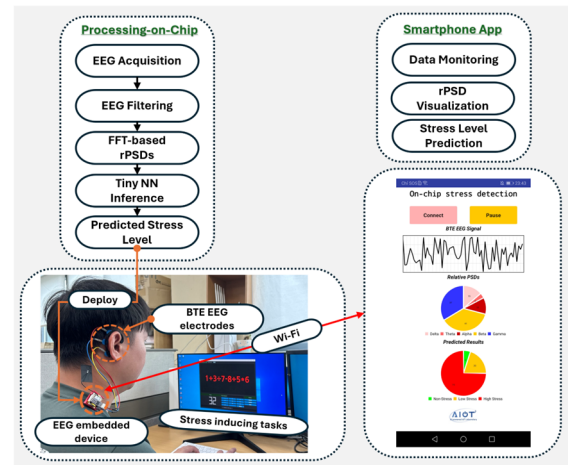


Fig. 1. Proposed system overview.

classifier, focusing on changes in EEG power spectral density within the theta and alpha bands, achieving classification accuracies of 79% and 78%, respectively [7]. Another study [8] assessed frontal EEG using a single-channel approach, reporting accuracy levels ranging from 65% to 75%. Additionally, Priya et al. [9] compared the performance of SVM and K-Nearest Neighbors (KNN) in stress classification.

The prevailing method for stress detection currently relies on machine learning (ML) and monitoring EEG signals from the scalp. However, this setup is often bulky and intrusive for everyday use. Moreover, the data processing unit typically relies on large hardware like cloud servers and personal computers, receiving EEG signals wirelessly from wearable sensors. This leads to a cumbersome system with limited mobility, high latency, and power consumption due to wireless connections. The BTE EEG presents a promising alternative, offering a more convenient and practical solution compared to scalp EEG. Recent advancements in miniature ML suggest that on-device ML-based stress detection could overcome the aforementioned limitations. This study introduces a novel BTE EEG-based multi-level stress detection system, employing a wearable embedded device running built-in neural network models.

II. MATERIALS AND METHODS

A. System Overview

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. NRF-2019R1A2C1089139).

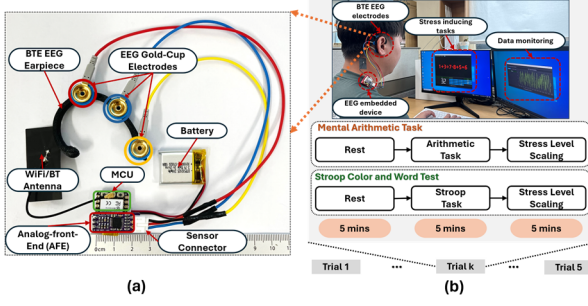


Fig. 2. (a) Proposed embedded device, (b) Experimental protocol.

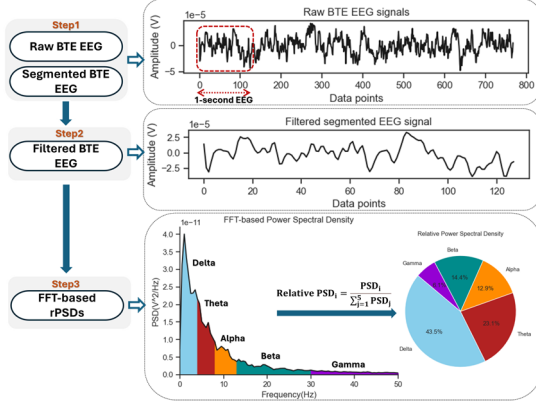


Fig. 3. Feature extraction procedure.

Our multi-level stress detection system has two parts: the on-chip BTE-EEG embedded device and a smartphone app (see Fig. 1). The wearable device collects EEG signals from behind the ear, filters them to keep signals between 1Hz and 50Hz, and extracts relative PSD features over five frequency bands: delta (1-4Hz), theta (4-8Hz), alpha (8-14Hz), beta (14-30Hz), and gamma (30-50Hz) using FFT. These features are input into a tiny neural network for stress classification. All these steps are performed directly on our embedded device (on-chip processing). The processed data is sent to a smartphone app via Wi-Fi for easy and quick stress level monitoring.

B. BTE-EEG Embedded Device

Fig. 2 (a) depicts the components of our BTE EEG-embedded device. The device has a pleasant, 3D-printed earpiece made of flexible plastic that fits the wearer's ear. Three EEG gold-cup electrodes are attached: the yellow one is used as a reference, while the blue and red ones gather EEG data from behind the ear. These impulses are sent through the BioAmp EXG Pill [10], a tiny device that captures high-quality EEG data. The processed signals are then handled by the ESP32-C3, a sophisticated IoT development board with WiFi and Bluetooth capabilities that includes a 32-bit RISC-V CPU and an FPU for precise computation. This makes the ESP32-C3 perfect for signal processing and executing tiny neural networks on the device. The processed EEG results are communicated to a smartphone app via a Wi-Fi antenna, enabling easy and rapid monitoring of the individual's stress levels.

C. Experimental Protocol

This study included 10 students from Pukyong National University, with an average age of 31 ± 6 years. EEG signals

TABLE I. THE PROPOSED NN'S ARCHITECTURE

Layer	Output Shape	Activation	Param #
Input	(None, 5)	---	0
Dense	(None, 128)	ReLU	768
Batch Normalization	(None, 128)	---	512
Dropout (0.5)	(None, 128)	---	0
Dense	(None, 128)	ReLU	16512
Batch Normalization	(None, 128)	---	512
Dropout (0.5)	(None, 128)	---	0
Dense	(None, 64)	ReLU	8256
Batch Normalization	(None, 64)	---	256
Dropout (0.3)	(None, 64)	---	0
Output	(None, 3)	SoftMax	195
Total params:	Trainable params:	Non-trainable params:	
26,011	25,499	512	

were acquired while the subjects were under stress caused by the Stroop task and the Mental Arithmetic task. Each stress-inducing task consisted of 5 minutes of relaxation followed by 5 minutes of the stressor, with 5 minutes set aside for participants to rate their stress levels on a scale of 0–2. As indicated in Fig. 2 (b), 0 represents no stress, 1 denotes low stress, and 2 indicates high stress. The experiment gained ethical approval from the Pukyong National University Institutional Review Board, reference number 1041386-202003-HR-11-02.

D. FFT-based Relative PSD Feature Extraction

The relative PSD (rPSD) features in this study are key EEG inputs for the stress classification model. They show the power distribution across different frequency bands. Using the FFT [11], EEG data is divided into frequency components to calculate PSD for delta, theta, alpha, beta, and gamma bands. The rPSD features are derived by normalizing the power in each band by the total power across all bands, providing a dimensionless measure for easy comparison across subjects and conditions. The rPSD for each frequency band is calculated using this formula:

$$rPSD_j = \frac{PSD_j}{\sum_{j=1}^5 PSD_j}$$

where $rPSD_j$ is the relative PSD for the j -th frequency band, PSD_j is the PSD value for the j -th frequency band, and the sum in the denominator is the total PSD value across all five frequency bands ($j = 1$ to 5, corresponding to delta, theta, alpha, beta, and gamma).

E. Neural Network-based Multi-Level Stress Classification

The proposed neural network model for multilevel stress classification uses relative PSDs from five frequency bands as inputs. Its architecture includes an input layer, three fully connected hidden layers with 128, 128, and 64 neurons, respectively, using ReLU activation functions to capture complex patterns (described in Table I). To prevent

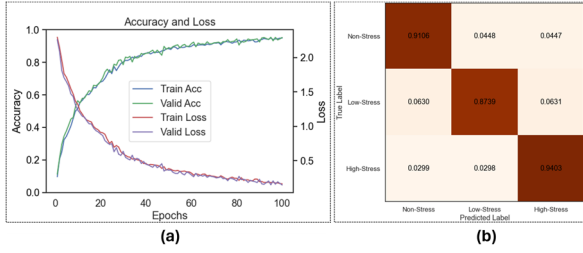


Fig. 4. (a) Accuracy and loss curves, and (b) Confusion matrix of the proposed NN.

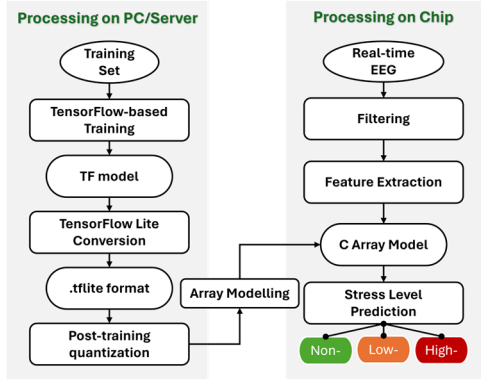


Fig. 5. Model conversion for on-chip deployment.

overfitting, Batch Normalization and Dropout layers (rates of 0.5 and 0.3) are added between the dense layers. The output layer uses a softmax activation function to generate class probabilities. This architecture balances complexity and regularization, optimized with the Adam optimizer at a learning rate of 0.001.

Additionally, our study employed a random forest (RF) model with 100 decision trees and a maximum depth of 10. RF is robust and efficiently captures complex patterns in physiological data, aiding in stress level evaluation. The feature importance analysis provided valuable insights into physiological stress biomarkers.

III. RESULTS AND DISCUSSION

A. Signal Preprocessing And Feature Extraction

Fig. 3 outlines the signal preprocessing and feature extraction process, comprising three key steps. Firstly, the raw BTE EEG signal from our device is segmented into 1-second segments, each containing 128 data points without overlap. These segments undergo band-pass filtering to eliminate noise and preserve the desired EEG signal within the 1Hz to 50Hz frequency range. Subsequently, the filtered EEG signal is transformed into relative PSD values across 5 frequency bands using FFT. These relative PSD features serve as inputs for stress level classification models in the following stages.

B. Multilevel Stress Classification

The neural network and random forest classifiers underwent LOOC validation, where one subject's data is reserved for testing and the rest for training/validation. Fig. 4 (a) displays accuracy and loss curves of the neural network on subject 6's data post 100 epochs with a batch size of 32. Evaluation metrics include accuracy, precision, recall, and F1-score. Accuracy measures overall

TABLE II. EVALUATION METRICS

Model	Acc	Pre	Recall	F1
Neural Network	0.9083	0.9087	0.9083	0.9081
Random Forest	0.8761	0.8763	0.8761	0.8761

TABLE III. DURATION (MS) OF PROCESSING-ON-CHIP TASKS

Processing-on-chip Task	Duration (ms)	Accuracy
Noise Filtering	18.95	---
Feature Extraction	395.41	---
Model Inference	Neural Network: 4.36 Random Forest: 64.12	0.9083 0.8761

correctness, precision indicates correct positive predictions, recall captures all actual positives, and F1-score balances precision and recall. Additionally, Fig. 4 (b) presents the confusion matrix for the neural network on the testing dataset. Table II compares both models' performance in stress classification, where the neural network outperforms the random forest with accuracy of 0.9083, precision of 0.9087, recall of 0.9083, and an F1-score of 0.9081, as opposed to the random forest's accuracy of 0.8761, precision of 0.8763, recall of 0.8761, and F1-score of 0.8761.

C. On-chip Neural Network Deployment

This study introduces an end-to-end on-chip processing method, where data collection, signal preprocessing, feature extraction, and classification model execution are all conducted directly and synchronously on the embedded device (on chip or MCU). Fig. 5 illustrates the process of building classification models using TensorFlow [12], converting them to the TFLite format with TensorFlow Lite conversion, applying post-training quantization for model optimization, and deploying the models for on-chip processing. Table III outlines the runtime durations of tasks performed on the chip and compares the execution speeds of the proposed NN and RF classifiers for multilevel stress classification. The proposed NN not only achieves higher accuracy in multilevel stress classification (90.83% compared to RF's 87.61%) but also demonstrates significantly faster model inference speeds (4.36ms compared to RF's 64.12ms).

IV. CONCLUSION

A novel stress prediction approach utilizing EEG data from BTE and TNNs has been introduced. This system, consisting of an EEG processing device and a smartphone app, emphasizes comfort and interference reduction. It filters out irrelevant signals, extracts features, and categorizes stress levels using a compact TNN. Crucially, all processing is on-chip, ensuring rapid, synchronized processing and averting data loss. Experimental findings demonstrated the effectiveness and viability of the proposed system for on-device multilevel stress detection.

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